Edugenie- Project Development Phases

5. Project Development Phase

The development phase involves implementing the planned modules and integrating them into a complete application.

5.1 Backend Development

FastAPI was selected as the backend framework because it provides a lightweight and efficient way to create REST APIs.

The backend handles:

- User requests.
- API routing.
- AI model communication.
- Response processing.
- Error handling.

5.2 Q&A Module

The Q&A module accepts a question from the user and sends it to the AI model.

Workflow

User Question

↓

FastAPI /qa

↓

Q&A Module

↓

Wikipedia

Gemini API

↓

Generated Answer

↓

Frontend

5.3 Explanation Module

This module converts complex topics into simple explanations suitable for learners.

The lightweight AI model can be used to provide concise educational explanations.

5.4 Quiz Module

The quiz module generates multiple-choice questions.

Each quiz contains:

- Question.
- Four options.
- Correct answer.
- Structured output.

The module also processes the AI response and handles JSON parsing errors.

5.5 Summary Module

The summary module accepts lengthy educational text and generates a concise version.

It removes unnecessary information while retaining the important concepts.

5.6 Learning Path Module

This module generates a personalized learning path.

The learning path can contain:

- Beginner topics.
- Intermediate topics.
- Advanced topics.
- Suggested timeline.
- Learning resources.
- Recommended sequence.

5.7 Frontend Development

The frontend was developed using HTML and CSS.

The interface contains:

- Task selection.
- Text input area.
- Submit button.
- Result display section.
- Responsive layout.

The frontend communicates with the FastAPI backend through HTTP requests.

